5

Concentration without independence

The approach to concentration inequalities we developed so far relies crucially on independence of random variables. We now pursue some alternative approaches to concentration, which are not based on independence. In Section 5.1, we demonstrate how to derive concentration from isoperimetric inequalities. We first do this on the example of the Euclidean sphere and then discuss other natural settings in Section 5.2.

In Section 5.3 we use concentration on the sphere to derive the classical Johnson-Lindenstrauss Lemma, a basic results about dimension reduction for high-dimensional data.

Section 5.4 introduces matrix concentration inequalities. We prove the matrix Bernstein's inequality, a remarkably general extension of the classical Bernstein inequality from Section 2.8 for random matrices. We then give two applications in Sections 5.5 and 5.6, extending our analysis for community detection and covariance estimation problems to sparse networks and fairly general distributions in $\mathbb{R}^n$.

5.1 Concentration of Lipschitz functions on the sphere

Consider a Gaussian random vector $X \sim N(0, I_n)$ and a function $f : \mathbb{R}^n \to \mathbb{R}$. When does the random variable $f(X)$ concentrate about its mean, i.e.

$$f(X) \approx \mathbb{E}\, f(X) \quad \text{with high probability?}$$

This question is easy for *linear functions* f. Indeed, in this case $f(X)$ has normal distribution, and it concentrates around its mean well (recall Exercise 3.3.3 and Proposition 2.1.2).

We now study concentration of *non-linear* functions $f(X)$ of random vectors X. We can not expect to have a good concentration for completely arbitrary f (why?). But if f does not oscillate too wildly, we might expect concentration. The concept of Lipschitz functions, which we introduce now, will help us to rigorously rule out functions that have wild oscillations.

5.1.1 Lipschitz functions

Definition 5.1.1 (Lipschitz functions)**.** Let (X, d_X) and (Y, d_Y) be metric spaces. A function $f : X \to Y$ is called *Lipschitz* if there exists $L \in \mathbb{R}$ such that

$$d_Y(f(u), f(v)) \le L \cdot d_X(u, v) \quad \text{for every } u, v \in X.$$

The infimum of all L in this definition is called the *Lipschitz norm* of f and is denoted $\|f\|_{\mathrm{Lip}}$.

In other words, Lipschitz functions may not blow up distances between points too much. Lipschitz functions with $\|f\|_{\mathrm{Lip}} \le 1$ are usually called *contractions*, since they may only shrink distances.

Lipschitz functions form an intermediate class between uniformly continuous and differentiable functions:

Exercise 5.1.2 (Continuity, differentiability, and Lipschitz functions)**.** ☕☕ Prove the following statements.

(a) Every Lipschitz function is uniformly continuous.
(b) Every differentiable function $f : \mathbb{R}^n \to \mathbb{R}$ is Lipschitz, and

$$\|f\|_{\mathrm{Lip}} \le \sup_{x \in \mathbb{R}^n} \|\nabla f(x)\|_2.$$

(c) Give an example of a non-Lipschitz but uniformly continuous function $f : [-1, 1] \to \mathbb{R}$.
(d) Give an example of a non-differentiable but Lipschitz function $f : [-1, 1] \to \mathbb{R}$.

Here are a few useful examples of Lipschitz functions on $\mathbb{R}^n$.

Exercise 5.1.3 (Linear functionals and norms as Lipschitz functions)**.** ☕☕ Prove the following statements.

(a) For a fixed $\theta \in \mathbb{R}^n$, the linear functional

$$f(x) = \langle x, \theta \rangle$$

is a Lipschitz function on $\mathbb{R}^n$, and $\|f\|_{\mathrm{Lip}} = \|\theta\|_2$.
(b) More generally, an $m \times n$ matrix A acting as a linear operator

$$A : (\mathbb{R}^n, \|\cdot\|_2) \to (\mathbb{R}^m, \|\cdot\|_2)$$

is Lipschitz, and $\|A\|_{\mathrm{Lip}} = \|A\|$.
(c) Any norm $f(x) = \|x\|$ on $(\mathbb{R}^n, \|\cdot\|_2)$ is a Lipschitz function. The Lipschitz norm of f is the smallest L that satisfies

$$\|x\| \le L\|x\|_2 \quad \text{for all } x \in \mathbb{R}^n.$$

5.1.2 Concentration via isoperimetric inequalities

The main result of this section is that any Lipschitz function on the Euclidean sphere $S^{n-1} = \{x \in \mathbb{R}^n : \|x\|_2 = 1\}$ concentrates well.

Theorem 5.1.4 (Concentration of Lipschitz functions on the sphere)**.** *Consider a random vector $X \sim \mathrm{Unif}(\sqrt{n}S^{n-1})$, i.e. X is uniformly distributed on the Euclidean sphere of radius $\sqrt{n}$. Consider a Lipschitz function*[1] $f : \sqrt{n}S^{n-1} \to \mathbb{R}$. *Then*

$$\big\| f(X) - \mathbb{E} f(X) \big\|_{\psi_2} \le C \|f\|_{\mathrm{Lip}}.$$

Using the definition of the sub-gaussian norm, the conclusion of Theorem 5.1.4 can be stated as follows: for every $t \ge 0$, we have

$$\mathbb{P}\Big\{ \big| f(X) - \mathbb{E} f(X) \big| \ge t \Big\} \le 2 \exp\Big(- \frac{ct^2}{\|f\|_{\mathrm{Lip}}^2} \Big).$$

Let us set out a strategy to prove Theorem 5.1.4. We already proved it for linear functions. Indeed, Theorem 3.4.6 states that $X \sim \mathrm{Unif}(\sqrt{n}S^{n-1})$ is a sub-gaussian random vector, and this by definition means that any linear function of X is a sub-gaussian random variable.

To prove Theorem 5.1.4 in full generality, we will argue that any non-linear Lipschitz function must concentrate at least as strongly as a linear function. To show this, instead of comparing non-linear to linear functions directly, we will compare the areas of their *sub-level sets* – the subsets of the sphere of the form $\{x : f(x) \le a\}$. The sub-level sets of linear functions are obviously the spherical caps. We can compare the areas of general sets and spherical caps using a remarkable geometric principle – an *isoperimetric inequality.*

The most familiar form of an isoperimetric inequality applies to subsets of $\mathbb{R}^3$ (and also in $\mathbb{R}^n$):

Theorem 5.1.5 (Isoperimetric inequality on $\mathbb{R}^n$)**.** *Among all subsets $A \subset \mathbb{R}^n$ with given volume, the Euclidean balls have minimal area. Moreover, for any $\varepsilon > 0$, the Euclidean balls minimize the volume of the ε-neighborhood of A, defined as*[2]

$$A_\varepsilon := \big\{ x \in \mathbb{R}^n : \exists y \in A \text{ such that } \|x - y\|_2 \le \varepsilon \big\} = A + \varepsilon B_2^n.$$

Figure 5.1 illustrates the isoperimetric inequality. Note that the "moreover" part of Theorem 5.1.5 implies the first part: to see this, let $\varepsilon \to 0$.

A similar isoperimetric inequality holds for subsets of the sphere S^{n-1}, and in this case the minimizers are the *spherical caps* – the neighborhoods of a single point.[3] To state this principle, we denote by σ_{n-1} the normalized area on the sphere S^{n-1} (i.e. the $n-1$-dimensional Lebesgue measure).

[1] This theorem is valid for both the geodesic metric on the sphere (where $d(x, y)$ is the length of the shortest arc connecting x and y) and the Euclidean metric $d(x, y) = \|x - y\|_2$. We will prove the theorem for the Euclidean metric; Exercise 5.1.11 extends it to geodesic metric.

[2] Here we used the notation for Minkowski sum introduced in Definintion 4.2.11.

[3] More formally, a closed spherical cap centered at a point $a \in S^{n-1}$ and with radius ε can be defined as $C(a, \varepsilon) = \{x \in S^{n-1} : \|x - a\|_2 \le \varepsilon\}$.

Figure 5.1 Isoperimetric inequality in $\mathbb{R}^n$ states that among all sets A of given volume, the Euclidean balls minimize the volume of the ε-neighborhood A_ε.

Theorem 5.1.6 (Isoperimetric inequality on the sphere)**.** *Let $\varepsilon > 0$. Then, among all sets $A \subset S^{n-1}$ with given area $\sigma_{n-1}(A)$, the spherical caps minimize the area of the neighborhood $\sigma_{n-1}(A_\varepsilon)$, where*

$$A_\varepsilon := \left\{ x \in S^{n-1} : \ \exists y \in A \text{ such that } \|x - y\|_2 \le \varepsilon \right\}.$$

We do not prove isoperimetric inequalities (Theorems 5.1.5 ans 5.1.6) in this book; the bibliography notes for this chapter refer to several known proofs of these results.

5.1.3 Blow-up of sets on the sphere

The isoperimetric inequality implies a remarkable phenomenon that may sound counter-intuitive: if a set A makes up at least a *half* of the sphere (in terms of area) then the neighborhood A_ε will make up *most* of the sphere. We now state and prove this "blow-up" phenomenon, and then try to explain it heuristically. In view of Theorem 5.1.4, it will be convenient for us to work with the sphere of radius $\sqrt{n}$ rather than the unit sphere.

Lemma 5.1.7 (Blow-up)**.** *Let A be a subset of the sphere $\sqrt{n}S^{n-1}$, and let σ denote the normalized area on that sphere. If $\sigma(A) \ge 1/2$, then,*[4] *for every $t \ge 0$,*

$$\sigma(A_t) \ge 1 - 2\exp(-ct^2).$$

Proof Consider the hemisphere defined by the first coordinate:

$$H := \left\{ x \in \sqrt{n}S^{n-1} : \ x_1 \le 0 \right\}.$$

By assumption, $\sigma(A) \ge 1/2 = \sigma(H)$, so the isoperimetric inequality (Theorem 5.1.6) implies that

$$\sigma(A_t) \ge \sigma(H_t). \tag{5.1}$$

The neighborhood H_t of the hemisphere H is a spherical cap, and we could

[4] Here the neighborhood A_t of a set A is defined in the same way as before, that is $A_t := \left\{ x \in \sqrt{n}S^{n-1} : \ \exists y \in A \text{ such that } \|x - y\|_2 \le t \right\}$.

compute its area by a direct calculation. It is, however, easier to use Theorem 3.4.6 instead, which states a random vector

$$X \sim \text{Unif}(\sqrt{n}S^{n-1})$$

is sub-gaussian, and $\|X\|_{\psi_2} \le C$. Since σ is the uniform probability measure on the sphere, it follows that

$$\sigma(H_t) = \mathbb{P}\left\{X \in H_t\right\}.$$

Now, the definition of the neighborhood implies that

$$H_t \supset \left\{x \in \sqrt{n}S^{n-1} :\ x_1 \le t/\sqrt{2}\right\}. \tag{5.2}$$

(Check this – see Exercise 5.1.8.) Thus

$$\sigma(H_t) \ge \mathbb{P}\left\{X_1 \le t/\sqrt{2}\right\} \ge 1 - 2\exp(-ct^2).$$

The last inequality holds because $\|X_1\|_{\psi_2} \le \|X\|_{\psi_2} \le C$. In view of (5.1), the lemma is proved. □

Exercise 5.1.8. ☕☕ Prove inclusion (5.2).

The number 1/2 for the area in Lemma 5.1.7 was rather arbitrary. As the next exercise shows, 1/2 it can be changed to any constant and even to an exponentially small quantity.

Exercise 5.1.9 (Blow-up of exponentially small sets)**.** ☕☕☕ Let A be a subset of the sphere $\sqrt{n}S^{n-1}$ such that

$$\sigma(A) > 2\exp(-cs^2) \quad \text{for some } s > 0.$$

(a) Prove that $\sigma(A_s) > 1/2$.
(b) Deduce from this that for any $t \ge s$,

$$\sigma(A_{2t}) \ge 1 - 2\exp(-ct^2).$$

Here $c > 0$ is the absolute constant from Lemma 5.1.7.

Hint: If the conclusion of the first part fails, the complement $B := (A_s)^c$ satisfies $\sigma(B) \ge 1/2$. Apply the blow-up Lemma 5.1.7 for B.

Remark 5.1.10 (Zero-one law)**.** The blow-up phenomenon we just saw may be quite counter-intuitive at first sight. How can an exponentially small set A in Exercise 5.1.9 undergo such a dramatic transition to an exponentially large set A_{2t} under such a small perturbation $2t$? (Remember that t can be much smaller than the radius $\sqrt{n}$ of the sphere.) However perplexing this may seem, this is a typical phenomenon in high dimensions. It is reminiscent of *zero-one laws* in probability theory, which basically state that events that are determined by many random variables tend to have probabilities either zero or one.

5.1.4 Proof of Theorem 5.1.4

Without loss of generality, we can assume that $\|f\|_{\mathrm{Lip}} = 1$. (Why?) Let M denote a median of $f(X)$, which by definition is a number satisfying[5]

$$\mathbb{P}\left\{f(X) \le M\right\} \ge \frac{1}{2} \quad \text{and} \quad \mathbb{P}\left\{f(X) \ge M\right\} \ge \frac{1}{2}.$$

Consider the sub-level set

$$A := \left\{x \in \sqrt{n}S^{n-1} : \ f(x) \le M\right\}.$$

Since $\mathbb{P}\left\{X \in A\right\} \ge 1/2$, Lemma 5.1.7 implies that

$$\mathbb{P}\left\{X \in A_t\right\} \ge 1 - 2\exp(-ct^2). \tag{5.3}$$

On the other hand, we claim that

$$\mathbb{P}\left\{X \in A_t\right\} \le \mathbb{P}\left\{f(X) \le M + t\right\}. \tag{5.4}$$

Indeed, if $X \in A_t$ then $\|X - y\|_2 \le t$ for some point $y \in A$. By definition, $f(y) \le M$. Since f Lipschitz with $\|f\|_{\mathrm{Lip}} = 1$, it follows that

$$f(X) \le f(y) + \|X - y\|_2 \le M + t.$$

This proves our claim (5.4).

Combining (5.3) and (5.4), we conclude that

$$\mathbb{P}\left\{f(X) \le M + t\right\} \ge 1 - 2\exp(-ct^2).$$

Repeating the argument for $-f$, we obtain a similar bound for the probability that $f(X) \ge M - t$. (Do this!) Combining the two, we obtain a similar bound for the probability that $|f(X) - M| \le t$, and conclude that

$$\|f(X) - M\|_{\psi_2} \le C.$$

It remains to replace the median M by the expectation $\mathbb{E} f$. This can be done easily by applying the Centering Lemma 2.6.8. (How?) The proof of Theorem 5.1.4 is now complete. □

Exercise 5.1.11 (Geodesic metric). ☕☕☕ We proved Theorem 5.1.4 for functions f that are Lipschitz with respect to the Euclidean metric $\|x - y\|_2$ on the sphere. Argue that the same result holds for the geodesic metric, which is the length of the shortest arc connecting x and y.

Exercise 5.1.12 (Concentration on the unit sphere). ☕ We stated Theorem 5.1.4 for the scaled sphere $\sqrt{n}S^{n-1}$. Deduce that a Lipschitz function f on the *unit* sphere S^{n-1} satisfies

$$\left\|f(X) - \mathbb{E} f(X)\right\|_{\psi_2} \le \frac{C\|f\|_{\mathrm{Lip}}}{\sqrt{n}}. \tag{5.5}$$

[5] The median may not be unique. However, for continuous and one-to-one functions f, the median is unique. (Check!)

where $X \sim \mathrm{Unif}(S^{n-1})$. Equivalently, for every $t \ge 0$, we have

$$\mathbb{P}\left\{ \left| f(X) - \mathbb{E} f(X) \right| \ge t \right\} \le 2 \exp\left(- \frac{cnt^2}{\|f\|_{\mathrm{Lip}}^2} \right) \tag{5.6}$$

In the geometric approach to concentration that we just presented, we first (a) proved a blow-up inequality (Lemma 5.1.7), then deduced (b) concentration about the median, and (c) replaced the median by expectation. The next two exercises shows that these steps can be reversed.

Exercise 5.1.13 (Concentration about expectation and median are equivalent)**.** ☕☕ Consider a random variable Z with median M. Show that

$$c\|Z - \mathbb{E} Z\|_{\psi_2} \le \|Z - M\|_{\psi_2} \le C\|Z - \mathbb{E} Z\|_{\psi_2},$$

where $c, C > 0$ are some absolute constants.

Hint: To prove the upper bound, assume that $\|Z - \mathbb{E} Z\|_{\psi_2} \le K$ and use the definition of the median to show that $|M - \mathbb{E} Z| \le CK$.

Exercise 5.1.14 (Concentration and the blow-up phenomenon are equivalent)**.** ☕☕☕ Consider a random vector X taking values in some metric space (T, d). Assume that there exists $K > 0$ such that

$$\left\| f(X) - \mathbb{E} f(X) \right\|_{\psi_2} \le K \|f\|_{\mathrm{Lip}}$$

for every Lipschitz function $f : T \to \mathbb{R}$. For a subset $A \subset T$, define $\sigma(A) := \mathbb{P}(X \in A)$. (Then σ is a probability measure on T.) Show that if $\sigma(A) \ge 1/2$ then,[6] for every $t \ge 0$,

$$\sigma(A_t) \ge 1 - 2\exp(-ct^2/K^2)$$

where $c > 0$ is an absolute constant.

Hint: First replace the expectation by the median. Then apply the assumption for the function $f(x) := \mathrm{dist}(x, A) = \inf\{d(x, y) : \ y \in A\}$ whose median is zero.

Exercise 5.1.15 (Exponential set of mutually almost orthogonal points)**.** ☕☕☕ From linear algebra, we know that any set of orthonormal vectors in $\mathbb{R}^n$ must contain at most n vectors. However, if we allow the vectors to be almost orthogonal, there can be *exponentially many* of them! Prove this counterintuitive fact as follows. Fix $\varepsilon \in (0, 1)$. Show that there exists a set $\{x_1, \ldots, x_N\}$ of unit vectors in $\mathbb{R}^n$ which are mutually almost orthogonal:

$$|\langle x_i, x_j \rangle| \le \varepsilon \quad \text{for all } i \ne j,$$

and the set is *exponentially large* in n:

$$N \ge \exp\big(c(\varepsilon) n\big).$$

Hint: Construct the points $x_i \in S^{n-1}$ one at a time. Note that the set of points on the sphere that are *not* almost orthogonal with a given point x_0 form two spherical caps. Show that the normalized area of the cap is exponentially small.

[6] Here the neighborhood A_t of a set A is defined in the same way as before, that is $A_t := \big\{x \in T : \ \exists y \in A \text{ such that } d(x, y) \le t\big\}$.

5.2 Concentration on other metric measure spaces

In this section, we extend the concentration for the sphere to other spaces. To do this, note that our proof of Theorem 5.1.4. was based on two main ingredients:

(a) an isoperimetric inequality;
(b) a blow-up of the minimizers for the isoperimetric inequality.

These two ingredients are not special to the sphere. Many other metric measure spaces satisfy (a) and (b), too, and thus concentration can be proved in such spaces as well. We will discuss two such examples, which lead to Gaussian concentration in $\mathbb{R}^n$ and concentration on the Hamming cube, and then we will mention a few other situations where concentration can be shown.

5.2.1 Gaussian concentration

The classical isoperimetric inequality in $\mathbb{R}^n$, Theorem 5.1.5, holds not only with respect to the volume but also with respect to the *Gaussian measure* on $\mathbb{R}^n$. The Gaussian measure of a (Borel) set $A \subset \mathbb{R}^n$ is defined as[7]

$$\gamma_n(A) := \mathbb{P}\left\{X \in A\right\} = \frac{1}{(2\pi)^{n/2}} \int_A e^{-\|x\|_2^2/2}\, dx$$

where $X \sim N(0, I_n)$ is the standard normal random vector in $\mathbb{R}^n$.

Theorem 5.2.1 (Gaussian isoperimetric inequality)**.** *Let $\varepsilon > 0$. Then, among all sets $A \subset \mathbb{R}^n$ with fixed Gaussian measure $\gamma_n(A)$, the half spaces minimize the Gaussian measure of the neighborhood $\gamma_n(A_\varepsilon)$.*

Using the method we developed for the sphere, we can deduce from Theorem 5.2.1 the following Gaussian concentration inequality.

Theorem 5.2.2 (Gaussian concentration)**.** *Consider a random vector $X \sim N(0, I_n)$ and a Lipschitz function $f : \mathbb{R}^n \to \mathbb{R}$ (with respect to the Euclidean metric). Then*

$$\big\|f(X) - \mathbb{E} f(X)\big\|_{\psi_2} \le C\|f\|_{\mathrm{Lip}}. \tag{5.7}$$

Exercise 5.2.3. ☕☕☕ Deduce Gaussian concentration inequality (Theorem 5.2.2) from Gaussian isoperimetric inequality (Theorem 5.2.1).
Hint: The ε-neighborhood of a half-space is still a half-space, and its Gaussian measure should be easy to compute.

Two partial cases of Theorem 5.2.2 should already be familiar:

(a) For *linear functions* f, Theorem 5.2.2 follows easily since the normal distribution $N(0, I_n)$ is sub-gaussian.
(b) For the *Euclidean norm* $f(x) = \|x\|_2$, Theorem 5.2.2 follows from Theorem 3.1.1.

[7] Recall the definition of the standard normal distribution in $\mathbb{R}^n$ from Section 3.3.2.

Exercise 5.2.4 (Replacing expectation by L^p norm). ☕☕☕ Prove that in the concentration results for sphere and Gauss space (Theorems 5.1.4 and 5.2.2), the expectation $\mathbb{E} f(X)$ can be replaced by the L^p norm $\left(\mathbb{E} f(X)^p\right)^{1/p}$ for any $p \geq 1$ and for any non-negative function f. The constants may depend on p.

5.2.2 Hamming cube

We saw how isoperimetry leads to concentration in two metric measure spaces, namely (a) the sphere S^{n-1} equipped with the Euclidean (or geodesic) metric and the uniform measure, and (b) $\mathbb{R}^n$ equipped with the Euclidean metric and Gaussian measure. A similar method yields concentration on many other metric measure spaces. One of them is the Hamming cube

$$\left(\{0,1\}^n, d, \mathbb{P}\right),$$

which we introduced in Definition 4.2.14. It will be convenient here to assume that $d(x, y)$ is the *normalized* Hamming distance, which is the fraction of the digits on which the binary strings x and y disagree, thus

$$d(x, y) = \frac{1}{n} \left|\{i : x_i \neq y_i\}\right|.$$

The measure $\mathbb{P}$ is the uniform probability measure on the Hamming cube, i.e.

$$\mathbb{P}(A) = \frac{|A|}{2^n} \quad \text{for any } A \subset \{0,1\}^n.$$

Theorem 5.2.5 (Concentration on the Hamming cube). *Consider a random vector $X \sim \mathrm{Unif}\{0,1\}^n$. (Thus, the coordinates of X are independent $\mathrm{Ber}(1/2)$ random variables.) Consider a function $f : \{0,1\}^n \to \mathbb{R}$. Then*

$$\|f(X) - \mathbb{E} f(X)\|_{\psi_2} \leq \frac{C\|f\|_{\mathrm{Lip}}}{\sqrt{n}}. \tag{5.8}$$

This result can be deduced from the isoperimetric inequality on the Hamming cube, whose minimizers are known to be the *Hamming balls* – the neighborhoods of single points with respect to the Hamming distance.

5.2.3 Symmetric group

The symmetric group S_n consists of all $n!$ permutations of n symbols, which we choose to be $\{1, \ldots, n\}$ to be specific. We can view the symmetric group as a metric measure space

$$(S_n, d, \mathbb{P}).$$

Here $d(\pi, \rho)$ is the normalized Hamming distance – the fraction of the symbols on which permutations π and ρ disagree:

$$d(\pi, \rho) = \frac{1}{n} \left|\{i : \pi(i) \neq \rho(i)\}\right|.$$

The measure $\mathbb{P}$ is the uniform probability measure on S_n, i.e.

$$\mathbb{P}(A) = \frac{|A|}{n!} \quad \text{for any } A \subset S_n.$$

Theorem 5.2.6 (Concentration on the symmetric group). *Consider a random permutation $X \sim \mathrm{Unif}(S_n)$ and a function $f : S_n \to \mathbb{R}$. Then the concentration inequality (5.8) holds.*

5.2.4 Riemannian manifolds with strictly positive curvature

A wide general class of examples with nice concentration properties is covered by the notion of a *Riemannian manifold*. Since we do not assume that the reader has necessary background in differential geometry, the rest of this section is optional.

Let (M, g) be a compact connected smooth Riemannian manifold. The canonical distance $d(x, y)$ on M is defined as the arclength (with respect to the Riemmanian tensor g) of a minimizing geodesic connecting x and y. The Riemannian manifold can be viewed as a metric measure space

$$(M, d, \mathbb{P})$$

where $\mathbb{P} = \frac{dv}{V}$ is the probability measure on M obtained from the Riemann volume element dv by dividing by V, the total volume of M.

Let $c(M)$ denote the infimum of the Ricci curvature tensor over all tangent vectors. Assuming that $c(M) > 0$, it can be proved using semigroup tools that

$$\|f(X) - \mathbb{E} f(X)\|_{\psi_2} \le \frac{C\|f\|_{\mathrm{Lip}}}{\sqrt{c(M)}} \tag{5.9}$$

for any Lipschitz function $f : M \to \mathbb{R}$.

To give an example, it is known that $c(S^{n-1}) = n - 1$. Thus (5.9) gives an alternative approach to concentration inequality (5.5) for the sphere S^{n-1}. We give several other examples next.

5.2.5 Special orthogonal group

The special orthogonal group $\mathrm{SO}(n)$ consists of all distance preserving linear transformations on $\mathbb{R}^n$. Equivalently, the elements of $\mathrm{SO}(n)$ are $n \times n$ orthogonal matrices whose determinant equals 1. We can view the special orthogonal group as a metric measure space

$$\left(\mathrm{SO}(n), \|\cdot\|_F, \mathbb{P}\right),$$

where the distance is the Frobenius norm[8] $\|A - B\|_F$ and $\mathbb{P}$ is the uniform probability measure on $\mathrm{SO}(n)$.

Theorem 5.2.7 (Concentration on the special orthogonal group). *Consider a random orthogonal matrix $X \sim \mathrm{Unif}(\mathrm{SO}(n))$ and a function $f : \mathrm{SO}(n) \to \mathbb{R}$. Then the concentration inequality (5.8) holds.*

[8] The definition of Frobenius norm was given in Section 4.1.3.

This result can be deduced from concentration on general Riemannian manifolds, which we discussed in Section 5.2.4.

Remark 5.2.8 (Haar measure)**.** Here we do not go into detail about the formal definition of the uniform probability measure $\mathbb{P}$ on $\mathrm{SO}(n)$. Let us just mention for an interested reader that $\mathbb{P}$ is the *Haar measure* on $\mathrm{SO}(n)$ – the unique probability measure that is invariant under the action on the group.[9]

One can explicitly construct a random orthogonal matrix $X \sim \mathrm{Unif}(SO(n))$ in several ways. For example, we can make it from an $n \times n$ Gaussian random matrix G with $N(0,1)$ independent entries. Indeed, consider the singular value decomposition

$$G = U \Sigma V^{\mathsf{T}}.$$

Then the matrix of left singular vectors $X := U$ is uniformly distributed in $\mathrm{SO}(n)$. One can then define Haar measure μ on $SO(n)$ by setting

$$\mu(A) := \mathbb{P}\{X \in A\} \quad \text{for } A \subset \mathrm{SO}(n).$$

(The rotation invariance should be straightforward – check it!)

5.2.6 Grassmannian

The Grassmannian, or Grassmann manifold $G_{n,m}$ consists of all m-dimensional subspaces of $\mathbb{R}^n$. In the special case where $m = 1$, the Grassman manifold can be identified with the sphere S^{n-1} (how?), so the concentration result we are about to state will include the concentration on the sphere as a special case.

We can view the Grassmann manifold as a metric measure space

$$(G_{n,m}, d, \mathbb{P}).$$

The distance between subspaces E and F can be defined as the operator norm[10]

$$d(E, F) = \|P_E - P_F\|$$

where P_E and P_F are the orthogonal projections onto E and F, respectively.

The probability $\mathbb{P}$ is, like before, the uniform (Haar) probability measure on $G_{n,m}$. This measure allows us to talk about *random m-dimensional subspaces of $\mathbb{R}^n$*

$$E \sim \mathrm{Unif}(G_{n,m}),$$

Alternatively, a random subspace E (and thus the Haar measure on the Grassmannian) can be constructed by computing the column span (i.e. the image) of a random $n \times m$ Gaussian random matrix G with i.i.d. $N(0,1)$ entries. (The rotation invariance should be straightforward – check it!)

[9] A measure μ on $\mathrm{SO}(n)$ is rotation invariant if for any measurable set $E \subset \mathrm{SO}(n)$ and any $T \in \mathrm{SO}(n)$, one has $\mu(E) = \mu(T(E))$.

[10] The operator norm was introduced in Section 4.1.2.

Theorem 5.2.9 (Concentration on the Grassmannian)**.** *Consider a random subspace $X \sim \mathrm{Unif}(G_{n,m})$ and a function $f : G_{n,m} \to \mathbb{R}$. Then the concentration inequality* (5.8) *holds.*

This result can be deduced from concentration on the special orthogonal group from Section 5.2.5. (For the interested reader let us mention how: one can express that Grassmannian as the quotient $G_{n,m} = SO(n)/(SO_m \times SO_{n-m})$ and use the fact that concentration passes on to quotients.)

5.2.7 Continuous cube and Euclidean ball

Similar concentration inequalities can be proved for the unit Euclidean cube $[0,1]^n$ and the Euclidean ball[11] $\sqrt{n}B_2^n$, both equipped with Euclidean distance and uniform probability measures. This can be deduce then from Gaussian concentration by *pushing forward* the Gaussian measure to the uniform measures on the ball and the cube, respectively. We state these two theorems and prove them in a few exercises.

Theorem 5.2.10 (Concentration on the continuous cube)**.** *Consider a random vector $X \sim \mathrm{Unif}([0,1]^n)$. (Thus, the coordinates of X are independent random variables uniformly distributed on $[0,1]$.) Consider a Lipschitz function $f : [0,1]^n \to \mathbb{R}$. (The Lipschitz norm is with respect to the Euclidean distance.) Then the concentration inequality* (5.7) *holds.*

Exercise 5.2.11 (Pushing forward Gaussian to uniform distribution)**.** Let $\Phi(x)$ denote the cumulative distribution function of the standard normal distribution $N(0,1)$. Consider a random vector $Z = (Z_1, \ldots, Z_n) \sim N(0, I_n)$. Check that

$$\phi(Z) := \Big(\Phi(Z_1), \ldots, \Phi(Z_n)\Big) \sim \mathrm{Unif}([0,1]^n).$$

Exercise 5.2.12 (Proving concentration on the continuous cube)**.** Expressing $X = \phi(Z)$ by the previous exercise, use Gaussian concentration to control the deviation of $f(\phi(Z))$ in terms of $\|f \circ \phi\|_{\mathrm{Lip}} \le \|f\|_{\mathrm{Lip}} \|\phi\|_{\mathrm{Lip}}$. Show that $\|\phi\|_{\mathrm{Lip}}$ is bounded by an absolute constant and complete the proof of Theorem 5.2.10.

Theorem 5.2.13 (Concentration on the Euclidean ball)**.** *Consider a random vector $X \sim \mathrm{Unif}(\sqrt{n}B_2^n)$. Consider a Lipschitz function $f : \sqrt{n}B_2^n \to \mathbb{R}$. (The Lipschitz norm is with respect to the Euclidean distance.) Then the concentration inequality* (5.7) *holds.*

Exercise 5.2.14 (Proving concentration on the Euclidean ball)**.** Use a similar method to prove Theorem 5.2.13. Define a function $\phi : \mathbb{R}^n \to \sqrt{n}B_2^n$ that pushes forward the Gaussian measure on $\mathbb{R}^n$ into the uniform measure on $\sqrt{n}B_2^n$, and check that ϕ has bounded Lipschitz norm.

[11] Recall that B_2^n denotes the unit Euclidean ball, i.e. $B_2^n = \{x \in \mathbb{R}^n : \|x\|_2 \le 1\}$, and $\sqrt{n}B_2^n$ is the Euclidean ball of radius $\sqrt{n}$.

5.2.8 Densities $e^{-U(x)}$

The push forward approach from last section can be used to obtain concentration for many other distributions in $\mathbb{R}^n$. In particular, suppose a random vector X has density of the form

$$f(x) = e^{-U(x)}$$

for some function $U : \mathbb{R}^n \to \mathbb{R}$. As an example, if $X \sim N(0, I_n)$ then the normal density (3.4) gives $U(x) = \|x\|_2^2 + c$ where c is a constant (that depends on n but not x), and Gaussian concentration holds for X.

Now, if U is a general function whose curvature is at least like $\|x\|_2^2$, then we should expect at least Gaussian concentration. This is exactly what the next theorem states. The curvature of U is measured with the help of the *Hessian* $\operatorname{Hess} U(x)$, which by definition is the $n \times n$ symmetric matrix whose (i, j)-th entry equals $\partial^2 U / \partial x_i \partial x_j$.

Theorem 5.2.15. *Consider a random vector X in $\mathbb{R}^n$ whose density has the form $f(x) = e^{-U(x)}$ for some function $U : \mathbb{R}^n \to \mathbb{R}$. Assume there exists $\kappa > 0$ such that*[12]

$$\operatorname{Hess} U(x) \succeq \kappa I_n \quad \textit{for all } x \in \mathbb{R}^n.$$

Then any Lipschitz function $f : \mathbb{R}^n \to \mathbb{R}$ satisfies

$$\|f(X) - \mathbb{E} f(X)\|_{\psi_2} \le \frac{C\|f\|_{\mathrm{Lip}}}{\sqrt{\kappa}}.$$

Note a similarity of this theorem with the concentration inequality (5.9) for Riemannian manifolds. Both of them can be proved using semigroup tools, which we do not present in this book.

5.2.9 Random vectors with independent bounded coordinates

There is a remarkable partial generalization of Theorem 5.2.10 for random vectors $X = (X_1, \ldots, X_n)$ whose coordinates are independent and have *arbitrary* bounded distributions. By scaling, there is no loss of generality to assume that $|X_i| \le 1$, but we no longer require that X_i be *uniformly* distributed.

Theorem 5.2.16 (Talagrand's concentration inequality)**.** *Consider a random vector $X = (X_1, \ldots, X_n)$ whose coordinates are independent and satisfy*

$$|X_i| \le 1 \quad \textit{almost surely.}$$

Then concentration inequality (5.7) holds for any convex *Lipschitz function $f : [-1, 1]^n \to \mathbb{R}$.*

In particular, Talagrand's concentration inequality holds for any *norm* on $\mathbb{R}^n$. We do not prove this result here.

[12] The matrix inequality here means $\operatorname{Hess} U(x) - \kappa I_n$ is a symmetric positive semidefinite matrix.

5.3 Application: Johnson-Lindenstrauss Lemma

Suppose we have N data points in $\mathbb{R}^n$ where n is very large. We would like to reduce dimension of the data without sacrificing too much of its geometry. The simplest form of dimension reduction is to project the data points onto a low-dimensional subspace

$$E \subset \mathbb{R}^n, \quad \dim(E) := m \ll n,$$

see Figure 5.2 for illustration. How shall we choose the subspace E, and how small its dimension m can be?

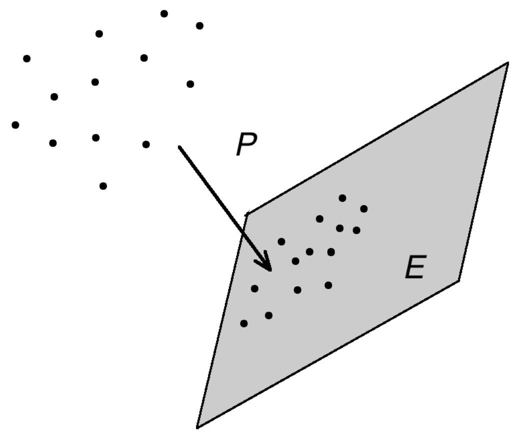

Figure 5.2 In Johnson-Lindenstrauss Lemma, the dimension of the data is reduced by projection onto a random low-dimensional subspace.

Johnson-Lindenstrauss Lemma below states that the geometry of data is well preserved if we choose E to be a *random subspace* of dimension

$$m \asymp \log N.$$

We already came across the notion of a random subspace in Section 5.2.6; let us recall it here. We say that E is a random m-dimensional subspace in $\mathbb{R}^n$ uniformly distributed in $G_{n,m}$, i.e.

$$E \sim \mathrm{Unif}(G_{n,m}),$$

if E is a random m-dimensional subspace of $\mathbb{R}^n$ whose distribution is rotation invariant, i.e.

$$\mathbb{P}\left\{E \in \mathcal{E}\right\} = \mathbb{P}\left\{U(E) \in \mathcal{E}\right\}$$

for any fixed subset $\mathcal{E} \subset G_{n,m}$ and $n \times n$ orthogonal matrix U.

Theorem 5.3.1 (Johnson-Lindenstrauss Lemma)**.** *Let $\mathcal{X}$ be a set of N points in $\mathbb{R}^n$ and $\varepsilon > 0$. Assume that*

$$m \geq (C/\varepsilon^2) \log N.$$

Consider a random m-dimensional subspace E in $\mathbb{R}^n$ uniformly distributed in $G_{n,m}$. Denote the orthogonal projection onto E by P. Then, with probability at least $1-2\exp(-c\varepsilon^2 m)$, the scaled projection

$$Q := \sqrt{\frac{n}{m}}\, P$$

is an approximate isometry on $\mathcal{X}$:

$$(1-\varepsilon)\|x-y\|_2 \le \|Qx-Qy\|_2 \le (1+\varepsilon)\|x-y\|_2 \quad \text{for all } x,y \in \mathcal{X}. \tag{5.10}$$

The proof of Johnson-Lindenstrauss Lemma will be based on concentration of Lipschitz functions on the sphere, which we studied in Section 5.1. We use it to first examine how the random projection P acts on a *fixed* vector $x-y$, and then take *union bound* over all N^2 differences $x-y$.

Lemma 5.3.2 (Random projection)**.** *Let P be a projection in $\mathbb{R}^n$ onto a random m-dimensional subspace uniformly distributed in $G_{n,m}$. Let $z \in \mathbb{R}^n$ be a (fixed) point and $\varepsilon > 0$. Then:*

(a) $\left(\mathbb{E}\,\|Pz\|_2^2\right)^{1/2} = \sqrt{\frac{m}{n}}\,\|z\|_2$.

(b) With probability at least $1-2\exp(-c\varepsilon^2 m)$, we have

$$(1-\varepsilon)\sqrt{\frac{m}{n}}\,\|z\|_2 \le \|Pz\|_2 \le (1+\varepsilon)\sqrt{\frac{m}{n}}\,\|z\|_2.$$

Proof Without loss of generality, we may assume that $\|z\|_2 = 1$. (Why?) Next, we consider an equivalent model: instead of a random projection P acting on a fixed vector z, we consider a fixed projection P acting on a random vector z. Specifically, the distribution of $\|Pz\|_2$ will not change if we let P be fixed and

$$z \sim \mathrm{Unif}(S^{n-1}).$$

(Check this using rotation invariance!)

Using rotation invariance again, we may assume without loss of generality that P is the *coordinate projection* onto the first m coordinates in $\mathbb{R}^n$. Thus

$$\mathbb{E}\,\|Pz\|_2^2 = \mathbb{E}\sum_{i=1}^m z_i^2 = \sum_{i=1}^m \mathbb{E}\,z_i^2 = m\,\mathbb{E}\,z_1^2, \tag{5.11}$$

since the coordinates z_i of the random vector $z \sim \mathrm{Unif}(S^{n-1})$ are identically distributed. To compute $\mathbb{E}\,z_1^2$, note that

$$1 = \|z\|_2^2 = \sum_{i=1}^n z_i^2.$$

Taking expectations of both sides, we obtain

$$1 = \sum_{i=1}^n \mathbb{E}\,z_i^2 = n\,\mathbb{E}\,z_1^2,$$

which yields

$$\mathbb{E}\, z_1^2 = \frac{1}{n}.$$

Putting this into (5.11), we get

$$\mathbb{E}\, \|Pz\|_2^2 = \frac{m}{n}.$$

This proves the first part of the lemma.

The second part follows from concentration of Lipschitz functions on the sphere. Indeed,

$$f(x) := \|Px\|_2$$

is a Lipschitz function on S^{n-1}, and $\|f\|_{\mathrm{Lip}} = 1$. (Check!) Then concentration inequality (5.6) yields

$$\mathbb{P}\left\{ \left| \|Px\|_2 - \sqrt{\frac{m}{n}} \right| \geq t \right\} \leq 2\exp(-cnt^2).$$

(Here we also used Exercise 5.2.4 to replace $\mathbb{E}\,\|x\|_2$ by the $(\mathbb{E}\,\|x\|_2^2)^{1/2}$ in the concentration inequality.) Choosing $t := \varepsilon\sqrt{m/n}$, we complete the proof of the lemma. $\square$

Proof of Johnson-Lindenstrauss Lemma. Consider the difference set

$$\mathcal{X} - \mathcal{X} := \{x - y : \ x, y \in \mathcal{X}\}.$$

We would like to show that, with required probability, the inequality

$$(1-\varepsilon)\|z\|_2 \leq \|Qz\|_2 \leq (1+\varepsilon)\|z\|_2$$

holds for all $z \in \mathcal{X} - \mathcal{X}$. Since $Q = \sqrt{n/m}\, P$, this is inequality is equivalent to

$$(1-\varepsilon)\sqrt{\frac{m}{n}}\, \|z\|_2 \leq \|Pz\|_2 \leq (1+\varepsilon)\sqrt{\frac{m}{n}}\, \|z\|_2. \tag{5.12}$$

For any fixed z, Lemma 5.3.2 states that (5.12) holds with probability at least $1 - 2\exp(-c\varepsilon^2 m)$. It remains to take a union bound over $z \in \mathcal{X} - \mathcal{X}$. It follows that inequality (5.12) holds simultaneously for all $z \in \mathcal{X} - \mathcal{X}$, with probability at least

$$1 - |\mathcal{X} - \mathcal{X}| \cdot 2\exp(-c\varepsilon^2 m) \geq 1 - N^2 \cdot 2\exp(-c\varepsilon^2 m).$$

If $m \geq (C/\varepsilon^2)\log N$ then this probability is at least $1 - 2\exp(-c\varepsilon^2 m/2)$, as claimed. Johnson-Lindenstrauss Lemma is proved. $\square$

A remarkable feature of Johnson-Lindenstrauss lemma is dimension reduction map A is *non-adaptive*, it does not depend on the data. Note also that the ambient dimension n of the data plays no role in this result.

Exercise 5.3.3 (Johnson-Lindenstrauss with sub-gaussian matrices)**.** ☕☕☕ Let A be an $m \times n$ random matrix whose rows are independent, mean zero, sub-gaussian isotropic random vectors in $\mathbb{R}^n$. Show that the conclusion of Johnson-Lindenstrauss lemma holds for $Q = (1/\sqrt{m})A$.

Exercise 5.3.4 (Optimality of Johnson-Lindenstrauss)**.** ☕☕☕ Give an example of a set $\mathcal{X}$ of N points for which no scaled projection onto a subspace of dimension $m \ll \log N$ is an approximate isometry.

Hint: Set $\mathcal{X}$ be an orthogonal basis and show that the projected set defines a packing.

5.4 Matrix Bernstein's inequality

In this section, we show how to generalize concentration inequalities for sums of independent random variables $\sum X_i$ to sums of independent *random matrices*.

In this section, we prove a matrix version of Bernstein's inequality (Theorem 2.8.4), where the random variables X_i are replaced by random matrices, and the absolute value $|\cdot|$ is replaced by the operator norm $\|\cdot\|$. Remarkably, we will not require independence of entries, rows, or columns within each random matrix X_i.

Theorem 5.4.1 (Matrix Bernstein's inequality)**.** *Let $X_1, \ldots, X_N$ be independent, mean zero, $n \times n$ symmetric random matrices, such that $\|X_i\| \le K$ almost surely for all i. Then, for every $t \ge 0$, we have*

$$\mathbb{P}\Big\{ \Big\| \sum_{i=1}^N X_i \Big\| \ge t \Big\} \le 2n \exp\Big(-\frac{t^2/2}{\sigma^2 + Kt/3} \Big).$$

Here $\sigma^2 = \big\| \sum_{i=1}^N \mathbb{E} X_i^2 \big\|$ is the norm of the matrix variance of the sum.

In particular, we can express this bound as the mixture of sub-gaussian and sub-exponential tail, just like in the scalar Bernstein's inequality:

$$\mathbb{P}\Big\{ \Big\| \sum_{i=1}^N X_i \Big\| \ge t \Big\} \le 2n \exp\Big[-c \cdot \min\Big(\frac{t^2}{\sigma^2}, \frac{t}{K} \Big) \Big].$$

The proof of matrix Bernstein's inequality will be based on the following naïve idea. We try to repeat the classical argument based on moment generating functions (see Section 2.8), replacing scalars by matrices at each occurrence. In most of our argument this idea will work, except for one step that will be non-trivial to generalize. Before we dive into this argument, let us develop some *matrix calculus* which will allow us to treat matrices as scalars.

5.4.1 Matrix calculus

Throughout this section, we work with symmetric $n \times n$ matrices. As we know, the operation of addition $A + B$ generalizes painlessly from scalars to matrices. We need to be more careful with multiplication, since it is not commutative for matrices: in general, $AB \neq BA$. For this reason, matrix Bernstein's inequality is sometimes called *non-commutative* Bernstein's inequality. Functions of matrices are defined as follows.

Definition 5.4.2 (Functions of matrices)**.** Consider a function $f : \mathbb{R} \to \mathbb{R}$ and an $n \times n$ symmetric matrix X with eigenvalues λ_i and corresponding eigenvectors u_i. Recall that X can be represented as a spectral decomposition

$$X = \sum_{i=1}^{n} \lambda_i u_i u_i^{\mathsf{T}}.$$

Then define

$$f(X) := \sum_{i=1}^{n} f(\lambda_i) u_i u_i^{\mathsf{T}}.$$

In other words, to obtain the matrix $f(X)$ from X, we do not change the eigenvectors and apply f to the eigenvalues.

In the following exercise, we check that the definition of function of matrices agrees with the basic rules of matrix addition and multiplication.

Exercise 5.4.3 (Matrix polynomials and power series)**.** ☕☕

(a) Consider a polynomial

$$f(x) = a_0 + a_1 x + \cdots + a_p x^p.$$

Check that for a matrix X, we have

$$f(X) = a_0 I + a_1 X + \cdots + a_p X^p.$$

In the right side, we use the standard rules for matrix addition and multiplication, so in particular $X^p = X \cdots X$ (p times) there.

(b) Consider a convergent power series expansion of f about x_0:

$$f(x) = \sum_{k=1}^{\infty} a_k (x - x_0)^k.$$

Check that the series of matrix terms converges, and

$$f(X) = \sum_{k=1}^{\infty} a_k (X - x_0 I)^k.$$

As an example, for each $n \times n$ symmetric matrix X we have

$$e^X = I + X + \frac{X^2}{2!} + \frac{X^3}{3!} + \cdots$$

Just like scalars, matrices can be compared to each other. To do this, we define a *partial order* on the set of $n \times n$ symmetric matrices as follows.

Definition 5.4.4 (positive semidefinite order)**.** Recall that

$$X \succeq 0$$

if X is a symmetric positive semidefinite matrix. Equivalently, $X \succeq 0$ if X is symmetric and all eigenvalues of X satisfy $\lambda_i(X) \geq 0$. Next, we write

$$X \succeq Y \quad \text{and} \quad Y \preceq X$$

if $X - Y \succeq 0$.

Note that $\succeq$ is a partial, as opposed to total, order, since there are matrices for which neither $X \succeq Y$ nor $Y \succeq X$ hold. (Give an example!)

Exercise 5.4.5. ☕☕☕ Prove the following properties.

(a) $\|X\| \leq t$ if and only if $-tI \preceq X \preceq tI$.
(b) Let $f, g : \mathbb{R} \to \mathbb{R}$ be two functions. If $f(x) \leq g(x)$ for all $x \in \mathbb{R}$ satisfying $|x| \leq K$, then $f(X) \preceq g(X)$ for all X satisfying $\|X\| \leq K$.
(c) Let $f : \mathbb{R} \to \mathbb{R}$ be an increasing function and X, Y are commuting matrices. Then $X \preceq Y$ implies $f(X) \preceq f(Y)$.
(d) Give an example showing that property (c) may fail for non-commuting matrices.

 Hint: Find 2×2 matrices such that $0 \preceq X \preceq Y$ but $X^2 \not\preceq Y^2$.
(e) In the following parts of the exercise, we develop weaker versions of property (c) that hold for arbitrary, not necessarily commuting, matrices. First, show that $X \preceq Y$ always implies $\operatorname{tr} f(X) \leq \operatorname{tr} f(Y)$ for any increasing function $f : \mathbb{R} \to \mathbb{R}$.

 Hint: Using Courant-Fisher's min-max principle (4.2), show that $\lambda_i(X) \leq \lambda_i(Y)$ for all i.
(f) Show that $0 \preceq X \preceq Y$ implies $X^{-1} \succeq Y^{-1}$ if X is invertible.

 Hint: First consider the case where one of the matrices is the identity. Next, multiply the inequality $X \preceq Y$ by $Y^{-1/2}$ on the left and on the right.
(g) Show that $0 \preceq X \preceq Y$ implies $\log X \preceq \log Y$.

 Hint: Check and use the identity $\log x = \int_0^\infty \left(\frac{1}{1+t} - \frac{1}{x+t}\right) dt$ and property (f).

5.4.2 Trace inequalities

So far, our attempts to extend scalar concepts for matrices have not met a considerable resistance. But this does not always go so smoothly. We already saw in Exercise 5.4.5 how the non-commutativity of the matrix product ($AB \neq BA$) may cause scalar properties to fail for matrices. Here is one more such situation: the identity is $e^{x+y} = e^x e^y$ holds for scalars but fails for matrices.

Exercise 5.4.6. ☕☕☕ Let X and Y be $n \times n$ symmetric matrices.

(a) Show that if the matrices commute, i.e. $XY = YX$, then

$$e^{X+Y} = e^X e^Y.$$

(b) Find an example of matrices X and Y such that

$$e^{X+Y} \neq e^X e^Y.$$

This is unfortunate for us, because we used the identity $e^{x+y} = e^x e^y$ in a crucial way in our approach to concentration of sums of random variables. Indeed, this identity allowed us to break the moment generating function $\mathbb{E}\exp(\lambda S)$ of the sum into the product of exponentials, see (2.6).

Nevertheless, there exists useful substitutes for the missing identity $e^{X+Y} = e^X e^Y$. We state two of them here without proof; they belong to the rich family of *trace inequalities.*

Theorem 5.4.7 (Golden-Thompson inequality)**.** *For any $n \times n$ symmetric matrices A and B, we have*

$$\operatorname{tr}(e^{A+B}) \leq \operatorname{tr}(e^A e^B).$$

Unfortunately, Goldon-Thpmpson inequality does not hold for three or more matrices: in general, the inequality $\operatorname{tr}(e^{A+B+C}) \leq \operatorname{tr}(e^A e^B e^C)$ may fail.

Theorem 5.4.8 (Lieb's inequality)**.** *Let H be an $n \times n$ symmetric matrix. Define the function on matrices*

$$f(X) := \operatorname{tr}\exp(H + \log X).$$

Then f is concave on the space on positive definite $n \times n$ symmetric matrices.[13]

Note that in the scalar case where $n = 1$, the function f is linear and Lieb's inequality holds trivially.

A proof of matrix Bernstein's inequality can be based on either Golden-Thompson or Lieb's inequalities. We use Lieb's inequality, which we will now restate for random matrices. If X is a random matrix, then Lieb's and Jensen's inequalities imply that

$$\mathbb{E} f(X) \leq f(\mathbb{E} X).$$

(Why does Jensen's inequality hold for random matrices?) Applying this with $X = e^Z$, we obtain the following.

Lemma 5.4.9 (Lieb's inequality for random matrices)**.** *Let H be a fixed $n \times n$ symmetric matrix and Z be a random $n \times n$ symmetric matrix. Then*

$$\mathbb{E}\operatorname{tr}\exp(H + Z) \leq \operatorname{tr}\exp(H + \log \mathbb{E} e^Z).$$

5.4.3 Proof of matrix Bernstein's inequality

We are now ready to prove matrix Bernstein's inequality, Theorem 5.4.1, using Lieb's inequality.

[13] Concavity means that the inequality $f(\lambda X + (1-\lambda)Y) \geq \lambda f(X) + (1-\lambda) f(Y)$ holds for matrices X and Y, and for $\lambda \in [0, 1]$.

Step 1: Reduction to MGF. To bound the norm of the sum

$$S := \sum_{i=1}^{N} X_i,$$

we need to control the largest and smallest eigenvalues of S. We can do this separately. To put this formally, consider the largest eigenvalue

$$\lambda_{\max}(S) := \max_i \lambda_i(S)$$

and note that

$$\|S\| = \max_i |\lambda_i(S)| = \max\left(\lambda_{\max}(S),\, \lambda_{\max}(-S)\right). \tag{5.13}$$

To bound $\lambda_{\max}(S)$, we proceed with the method based on computing the moment generating function as we did in the scalar case, e.g. in Section 2.2. To this end, fix $\lambda \geq 0$ and use Markov's inequality to obtain

$$\mathbb{P}\left\{\lambda_{\max}(S) \geq t\right\} = \mathbb{P}\left\{e^{\lambda \cdot \lambda_{\max}(S)} \geq e^{\lambda t}\right\} \leq e^{-\lambda t}\, \mathbb{E}\, e^{\lambda \cdot \lambda_{\max}(S)}. \tag{5.14}$$

Since by Definition 5.4.2 the eigenvalues of $e^{\lambda S}$ are $e^{\lambda \cdot \lambda_i(S)}$, we have

$$E := \mathbb{E}\, e^{\lambda \cdot \lambda_{\max}(S)} = \mathbb{E}\, \lambda_{\max}(e^{\lambda S}).$$

Since the eigenvalues of $e^{\lambda S}$ are all positive, the maximal eigenvalue of $e^{\lambda S}$ is bounded by the sum of all eigenvalues, the trace of $e^{\lambda S}$, which leads to

$$E \leq \mathbb{E} \operatorname{tr} e^{\lambda S}.$$

Step 2: Application of Lieb's inequality. To prepare for an application of Lieb's inequality (Lemma 5.4.9), let us separate the last term from the sum S:

$$E \leq \mathbb{E} \operatorname{tr} \exp\left[\sum_{i=1}^{N-1} \lambda X_i + \lambda X_N\right].$$

Condition on $(X_i)_{i=1}^{N-1}$ and apply Lemma 5.4.9 for the fixed matrix $H := \sum_{i=1}^{N-1} \lambda X_i$ and the random matrix $Z := \lambda X_N$. We obtain

$$E \leq \mathbb{E} \operatorname{tr} \exp\left[\sum_{i=1}^{N-1} \lambda X_i + \log \mathbb{E}\, e^{\lambda X_N}\right].$$

(To be more specific here, we first apply Lemma 5.4.9 for the conditional expectation, and then take expectation of both sides using the law of total expectation.)

We continue in a similar way: separate the next term λX_{N-1} from the sum $\sum_{i=1}^{N-1} \lambda X_i$ and apply Lemma 5.4.9 again for $Z = \lambda X_{N-1}$. Repeating N times, we obtain

$$E \leq \operatorname{tr} \exp\left[\sum_{i=1}^{N} \log \mathbb{E}\, e^{\lambda X_i}\right]. \tag{5.15}$$

Step 3: MGF of the individual terms. It remains to bound the matrix-valued moment generating function $\mathbb{E}\, e^{\lambda X_i}$ for each term X_i. This is a standard task, and the argument will be similar to the scalar case.

Lemma 5.4.10 (Moment generating function). *Let X be an $n \times n$ symmetric mean zero random matrix such that $\|X\| \le K$ almost surely. Then*

$$\mathbb{E} \exp(\lambda X) \preceq \exp\left(g(\lambda)\, \mathbb{E}\, X^2\right) \quad \textit{where} \quad g(\lambda) = \frac{\lambda^2/2}{1 - |\lambda| K/3},$$

provided that $|\lambda| < 3/K$.

Proof First, note that we can bound the (scalar) exponential function by the first few terms of its Taylor's expansion as follows:

$$e^z \le 1 + z + \frac{1}{1 - |z|/3} \cdot \frac{z^2}{2}, \quad \text{if } |z| < 3.$$

(To get this inequality, write $e^z = 1 + z + z^2 \cdot \sum_{p=2}^{\infty} z^{p-2}/p!$ and use the bound $p! \ge 2 \cdot 3^{p-2}$.) Next, apply this inequality for $z = \lambda x$. If $|x| \le K$ and $|\lambda| < 3/K$ then we obtain

$$e^{\lambda x} \le 1 + \lambda x + g(\lambda) x^2,$$

where $g(\lambda)$ is the function in the statement of the lemma.

Finally, we can transfer this inequality from scalars to matrices using part (b) of Exercise 5.4.5. We obtain that if $\|X\| \le K$ and $|\lambda| < 3/K$, then

$$e^{\lambda X} \preceq I + \lambda X + g(\lambda) X^2.$$

Take expectation of both sides and use the assumption that $\mathbb{E}\, X = 0$ to obtain

$$\mathbb{E}\, e^{\lambda X} \preceq I + g(\lambda)\, \mathbb{E}\, X^2.$$

To bound the right hand side, we may use the inequality $1 + z \le e^z$ which holds for all scalars z. Thus the inequality $I + Z \preceq e^Z$ holds for all matrices Z, and in particular for $Z = g(\lambda)\, \mathbb{E}\, X^2$. (Here we again refer to part (b) of Exercise 5.4.5.) This yields the conclusion of the lemma. □

Step 4: Completion of the proof. Let us return to bounding the quantity in (5.15). Using Lemma 5.4.10, we obtain

$$E \le \operatorname{tr} \exp\left[\sum_{i=1}^{N} \log \mathbb{E}\, e^{\lambda X_i}\right] \le \operatorname{tr} \exp\left[g(\lambda) Z\right], \quad \text{where} \quad Z := \sum_{i=1}^{N} \mathbb{E}\, X_i^2.$$

(Here we used Exercise 5.4.5 again: part (g) to take logarithms on both sides, and then part (e) to take traces of the exponential of both sides.)

Since the trace of $\exp\left[g(\lambda) Z\right]$ is a sum of n positive eigenvalues, it is bounded by n times the maximum eigenvalue, so

$$\begin{aligned} E &\le n \cdot \lambda_{\max}\left(\exp[g(\lambda) Z]\right) = n \cdot \exp\left[g(\lambda) \lambda_{\max}(Z)\right] \qquad \text{(why?)} \\ &= n \cdot \exp\left[g(\lambda) \|Z\|\right] \qquad \text{(since } Z \succeq 0\text{)} \\ &= n \cdot \exp\left[g(\lambda) \sigma^2\right] \qquad \text{(by definition of } \sigma \text{ in the theorem).} \end{aligned}$$

Plugging this bound for $E = \mathbb{E}\, e^{\lambda \cdot \lambda_{\max}(S)}$ into (5.14), we obtain

$$\mathbb{P}\left\{\lambda_{\max}(S) \geq t\right\} \leq n \cdot \exp\left[-\lambda t + g(\lambda)\sigma^2\right].$$

We obtained a bound that holds for any $\lambda > 0$ such that $|\lambda| < 3/K$, so we can minimize it in λ. Better yet, instead of computing the exact minimum (which might be a little too ugly), we can choose the following value: $\lambda = t/(\sigma^2 + Kt/3)$. Substituting it into the bound above and simplifying yields

$$\mathbb{P}\left\{\lambda_{\max}(S) \geq t\right\} \leq n \cdot \exp\left(-\frac{t^2/2}{\sigma^2 + Kt/3}\right).$$

Repeating the argument for $-S$ and combining the two bounds via (5.13), we complete the proof of Theorem 5.4.1. (Do this!) □

5.4.4 Matrix Khintchine's inequality

Matrix Bernstein's inequality gives a good *tail bound* on $\|\sum_{i=1}^N X_i\|$, and this in particular implies a non-trivial bound on the *expectation*:

Exercise 5.4.11 (Matrix Bernstein's inequality: expectation)**.** ☕☕☕ Let $X_1, \ldots, X_N$ be independent, mean zero, $n \times n$ symmetric random matrices, such that $\|X_i\| \leq K$ almost surely for all i. Deduce from Bernstein's inequality that

$$\mathbb{E}\left\|\sum_{i=1}^N X_i\right\| \lesssim \left\|\sum_{i=1}^N \mathbb{E}\, X_i^2\right\|^{1/2} \sqrt{1+\log n} + K(1+\log n).$$

Hint: Check that matrix Bernstein's inequality implies that $\|\sum_{i=1}^N X_i\| \lesssim \|\sum_{i=1}^N \mathbb{E}\, X_i^2\|^{1/2}\sqrt{\log n + u} + K(\log n + u)$ with probability at least $1 - 2e^{-u}$. Then use the integral identity from Lemma 1.2.1.

Note that in the scalar case where $n = 1$, a bound on the expectation is trivial. Indeed, in this case we have

$$\mathbb{E}\left|\sum_{i=1}^N X_i\right| \leq \left(\mathbb{E}\left|\sum_{i=1}^N X_i\right|^2\right)^{1/2} = \left(\sum_{i=1}^N \mathbb{E}\, X_i^2\right)^{1/2}.$$

where we used that the variance of a sum of independent random variables equals the sum of variances.

The techniques we developed in the proof of matrix Bernstein's inequality can be used to give matrix versions of other classical concentration inequalities. In the next two exercises, you will prove matrix versions of Hoeffding's inequality (Theorem 2.2.2) and Khintchine's inequality (Exercise 2.6.6).

Exercise 5.4.12 (Matrix Hoeffding's inequality)**.** ☕☕☕ Let $\varepsilon_1, \ldots, \varepsilon_n$ be independent symmetric Bernoulli random variables and let $A_1, \ldots, A_N$ be symmetric $n \times n$ matrices (deterministic). Prove that, for any $t \geq 0$, we have

$$\mathbb{P}\left\{\left\|\sum_{i=1}^N \varepsilon_i A_i\right\| \geq t\right\} \leq 2n \exp(-t^2/2\sigma^2),$$

where $\sigma^2 = \left\| \sum_{i=1}^N A_i^2 \right\|$.

Hint: Proceed like in the proof of Theorem 5.4.1. Instead of Lemma 5.4.10, check that $\mathbb{E}\exp(\lambda\varepsilon_i A_i) \preceq \exp(\lambda^2 A_i^2/2)$ just like in the proof of Hoeffding's inequality, Theorem 2.2.2.

From this, one can deduce a matrix version of Khintchine's inequality:

Exercise 5.4.13 (Matrix Khintchine's inequality)**.** ☕☕☕ Let $\varepsilon_1, \dots, \varepsilon_N$ be independent symmetric Bernoulli random variables and let $A_1, \dots, A_N$ be symmetric $n \times n$ matrices (deterministic).

(a) Prove that

$$\mathbb{E}\left\| \sum_{i=1}^N \varepsilon_i A_i \right\| \le C\sqrt{1+\log n}\left\| \sum_{i=1}^N A_i^2 \right\|^{1/2}.$$

(b) More generally, prove that for every $p \in [1, \infty)$ we have

$$\left(\mathbb{E}\left\| \sum_{i=1}^N \varepsilon_i A_i \right\|^p \right)^{1/p} \le C\sqrt{p+\log n}\left\| \sum_{i=1}^N A_i^2 \right\|^{1/2}.$$

The price of going from scalar to matrices is the pre-factor n in the probability bound in Theorem 5.4.1. This is a small price, considering that this factor becomes *logarithmic* in dimension n in the expectation bound of Exercises 5.4.11–5.4.13. The following example demonstrates that the logarithmic factor is needed in general.

Exercise 5.4.14 (Sharpness of matrix Bernstein's inequality)**.** ☕☕☕ Let X be an $n \times n$ random matrix that takes values $e_k e_k^\mathsf{T}$, $k = 1, \dots, n$, with probability $1/n$ each. (Here (e_k) denotes the standard basis in $\mathbb{R}^n$.) Let $X_1, \dots, X_N$ be independent copies of X. Consider the sum

$$S := \sum_{i=1}^N X_i,$$

which is a diagonal matrix.

(a) Show that the entry S_{ii} has the same distribution as the number of balls in i-th bin when N balls are thrown into n bins independently.

(b) Relating this to the classical coupon collector's problem, show that if $N \asymp n$ then[14]

$$\mathbb{E}\|S\| \asymp \frac{\log n}{\log\log n}.$$

Deduce that the bound in Exercise 5.4.11 would fail if the logarithmic factors were removed from it.

The following exercise extends matrix Bernstein's inequality by dropping both the symmetry and square assumption on the matrices X_i.

[14] Here we write $a_n \asymp b_n$ if there exist constants $c, C > 0$ such that $ca_n < b_n \le Ca_n$ for all n.

Exercise 5.4.15 (Matrix Bernstein's inequality for rectangular matrices). ☕☕☕
Let $X_1, \ldots, X_N$ be independent, mean zero, $m \times n$ random matrices, such that $\|X_i\| \le K$ almost surely for all i. Prove that for $t \ge 0$, we have

$$\mathbb{P}\left\{ \Big\| \sum_{i=1}^N X_i \Big\| \ge t \right\} \le 2(m+n) \exp\Big(- \frac{t^2/2}{\sigma^2 + Kt/3} \Big),$$

where

$$\sigma^2 = \max\Big(\Big\| \sum_{i=1}^N \mathbb{E}\, X_i^\mathsf{T} X_i \Big\|, \ \Big\| \sum_{i=1}^N \mathbb{E}\, X_i X_i^\mathsf{T} \Big\| \Big).$$

Hint: Apply matrix Bernstein's inequality (Theorem 5.4.1) for the sum of $(m+n) \times (m+n)$ symmetric matrices $\begin{bmatrix} 0 & X_i^\mathsf{T} \\ X_i & 0 \end{bmatrix}$.

5.5 Application: community detection in sparse networks

In Section 4.5, we analyzed a basic method for community detection in networks – the spectral clustering algorithm. We examined the performance of spectral clustering for the stochastic block model $G(n, p, q)$ with two communities, and we found how the communities can be identified with high accuracy and high probability (Theorem 4.5.6).

We now re-examine the performance of spectral clustering using matrix Bernstein's inequality. In the following two exercises, we find that spectral clustering actually works for *much sparser* networks than we knew before from Theorem 4.5.6.

Just like in Section 4.5, we denote by A the adjacency matrix of a random graph from $G(n, p, q)$, and we express A as

$$A = D + R$$

where $D = \mathbb{E}\, A$ is a deterministic matrix ("signal") and R is random ("noise"). As we know, the success of spectral clustering method hinges on the fact that the noise $\|R\|$ is small with high probability (recall (4.18)). In the following exercise, you will use Matrix Bernstein's inequality to derive a better bound on $\|R\|$.

Exercise 5.5.1 (Controlling the noise). ☕☕☕

(a) Represent the adjacency matrix A as a sum of independent random matrices

$$A = \sum_{1 \le i \le j \le n} Z_{ij}.$$

Make it so that each Z_{ij} encode the contribution of an edge between vertices i and j. Thus, the only non-zero entries of Z_{ij} should be (ij) and (ji), and they should be the same as in A.

(b) Apply matrix Bernstein's inequality to find that

$$\mathbb{E}\,\|R\| \lesssim \sqrt{d \log n} + \log n,$$

where $d = \frac{1}{2}(p+q)n$ is the *expected average degree* of the graph.

Exercise 5.5.2 (Spectral clustering for sparse networks). ☕☕☕ Use the bound from Exercise 5.5.1 to give better guarantees for the performance of spectral clustering than we had in Section 4.5. In particular, argue that spectral clustering works for *sparse networks*, as long as the average expected degrees satisfy

$$d \gg \log n.$$

5.6 Application: covariance estimation for general distributions

In Section 4.7, we saw how the covariance matrix of a sub-gaussian distribution in $\mathbb{R}^n$ can be accurately estimated using a sample of size $O(n)$. In this section, we remove the sub-gaussian requirement, and thus make covariance estimation possible for very general, in particular discrete, distributions. The price we pay is very small – just a logarithmic oversampling factor.

Like in Section 4.7, we estimate the the second moment matrix $\Sigma = \mathbb{E}\, XX^\mathsf{T}$ by its sample version

$$\Sigma_m = \frac{1}{m}\sum_{i=1}^m X_i X_i^\mathsf{T}.$$

Recall that if X has zero mean, then Σ is the covariance matrix of X and Σ_m is the sample covariance matrix of X.

Theorem 5.6.1 (General covariance estimation). *Let X be a random vector in $\mathbb{R}^n$, $n \ge 2$. Assume that for some $K \ge 1$,*

$$\|X\|_2 \le K\,(\mathbb{E}\,\|X\|_2^2)^{1/2} \quad \textit{almost surely.} \tag{5.16}$$

Then, for every positive integer m, we have

$$\mathbb{E}\,\|\Sigma_m - \Sigma\| \le C\left(\sqrt{\frac{K^2 n \log n}{m}} + \frac{K^2 n \log n}{m}\right)\|\Sigma\|.$$

Proof Before we start proving the bound, let us pause to note that $\mathbb{E}\,\|X\|_2^2 = \operatorname{tr}(\Sigma)$. (Check this like in the proof of Lemma 3.2.4.) So the assumption (5.16) becomes

$$\|X\|_2^2 \le K^2 \operatorname{tr}(\Sigma) \quad \text{almost surely.} \tag{5.17}$$

Apply the expectation version of matrix Bernstein's inequality (Exercise 5.4.11) for the sum of i.i.d. mean zero random matrices $X_i X_i^\mathsf{T} - \Sigma$ and get[15]

$$\mathbb{E}\,\|\Sigma_m - \Sigma\| = \frac{1}{m}\,\mathbb{E}\left\|\sum_{i=1}^m (X_i X_i^\mathsf{T} - \Sigma)\right\| \lesssim \frac{1}{m}\left(\sigma\sqrt{\log n} + M\log n\right) \tag{5.18}$$

[15] As usual, the notation $a \lesssim b$ hides absolute constant factors, i.e. it means that $a \le Cb$ where C is an absolute constant.

where

$$\sigma^2 = \Big\| \sum_{i=1}^m \mathbb{E}(X_i X_i^\mathsf{T} - \Sigma)^2 \Big\| = m \Big\| \mathbb{E}(XX^\mathsf{T} - \Sigma)^2 \Big\|$$

and M is any number chosen so that

$$\|XX^\mathsf{T} - \Sigma\| \le M \quad \text{almost surely.}$$

To complete the proof, it remains to bound σ^2 and M.

Let us start with σ^2. Expanding the square, we find that[16]

$$\mathbb{E}(XX^\mathsf{T} - \Sigma)^2 = \mathbb{E}(XX^\mathsf{T})^2 - \Sigma^2 \preceq \mathbb{E}(XX^\mathsf{T})^2. \tag{5.19}$$

Further, the assumption (5.17) gives

$$(XX^\mathsf{T})^2 = \|X\|^2 XX^\mathsf{T} \preceq K^2 \operatorname{tr}(\Sigma) XX^\mathsf{T}.$$

Taking expectation and recalling that $\mathbb{E}\, XX^\mathsf{T} = \Sigma$, we obtain

$$\mathbb{E}(XX^\mathsf{T})^2 \preceq K^2 \operatorname{tr}(\Sigma)\Sigma.$$

Substituting this bound into (5.19), we obtain a good bound on σ, namely

$$\sigma^2 \le K^2 m \operatorname{tr}(\Sigma)\|\Sigma\|.$$

Bounding M is simple: indeed,

$$\begin{aligned}
\|XX^\mathsf{T} - \Sigma\| &\le \|X\|_2^2 + \|\Sigma\| \quad \text{(by triangle inequality)} \\
&\le K^2 \operatorname{tr}(\Sigma) + \|\Sigma\| \quad \text{(by assumption (5.17))} \\
&\le 2K^2 \operatorname{tr}(\Sigma) =: M \quad \text{(since } \|\Sigma\| \le \operatorname{tr}(\Sigma) \text{ and } K \ge 1).
\end{aligned}$$

Substituting our bounds for σ and M into (5.18), we get

$$\mathbb{E}\, \|\Sigma_m - \Sigma\| \le \frac{1}{m}\Big(\sqrt{K^2 m \operatorname{tr}(\Sigma)\|\Sigma\|} \cdot \sqrt{\log n} + 2K^2 \operatorname{tr}(\Sigma) \cdot \log n \Big).$$

To complete the proof, use the inequality $\operatorname{tr}(\Sigma) \le n\|\Sigma\|$ and simplify the bound. □

Remark 5.6.2 (Sample complexity)**.** Theorem 5.6.1 implies that for any $\varepsilon \in (0,1)$, we are guaranteed to have covariance estimation with a good relative error,

$$\mathbb{E}\, \|\Sigma_m - \Sigma\| \le \varepsilon \|\Sigma\|, \tag{5.20}$$

if we take a sample of size

$$m \asymp \varepsilon^{-2} n \log n.$$

Compare this with the sample complexity $m \asymp \varepsilon^{-2} n$ for sub-gaussian distributions (recall Remark 4.7.2). We see that the price of dropping the sub-gaussian requirement turned out to be very small – it is just a logarithmic oversampling factor.

[16] Recall Definition 5.4.4 of the positive semidefinite order $\preceq$ used here.

Remark 5.6.3 (Lower-dimensional distributions)**.** At the end of the proof of Theorem 5.6.1, we used a crude bound $\operatorname{tr}(\Sigma) \le n\|\Sigma\|$. But we may chose not to do that, and instead get a bound in terms of the *intrinsic dimension*

$$r = \frac{\operatorname{tr}(\Sigma)}{\|\Sigma\|},$$

namely

$$\mathbb{E}\,\|\Sigma_m - \Sigma\| \le C\left(\sqrt{\frac{K^2 r \log n}{m}} + \frac{K^2 r \log n}{m}\right)\|\Sigma\|.$$

In particular, this stronger bound implies that a sample of size

$$m \asymp \varepsilon^{-2} r \log n$$

is sufficient to estimate the covariance matrix as in (5.20). Note that we always have $r \le n$ (why?), so the new bound is always as good as the one in Theorem 5.6.1. But for *approximately low dimensional* distributions – those that tend to concentrate near low-dimensional subspaces – we may have $r \ll n$, and in this case estimate covariance using a much smaller sample. We will return to this discussion in Section 7.6 where we introduce the notions of stable dimension and stable rank.

Exercise 5.6.4 (Tail bound)**.** ☕☕ Our argument also implies the following high-probability guarantee. Check that for any $u \ge 0$, we have

$$\|\Sigma_m - \Sigma\| \le C\left(\sqrt{\frac{K^2 r(\log n + u)}{m}} + \frac{K^2 r(\log n + u)}{m}\right)\|\Sigma\|$$

with probability at least $1 - 2e^{-u}$. Here $r = \operatorname{tr}(\Sigma)/\|\Sigma\| \le n$ as before.

Exercise 5.6.5 (Necessity of boundedness assumption)**.** ☕☕☕ Show that if the boundedness assumption (5.16) is removed from Theorem 5.6.1, the conclusion may fail in general.

Exercise 5.6.6 (Sampling from frames)**.** ☕☕ Consider an equal-norm tight frame[17] $(u_i)_{i=1}^N$ in $\mathbb{R}^n$. State and prove a result that shows that a random sample of

$$m \gtrsim n \log n$$

elements of (u_i) forms a frame with good frame bounds (as close to tight as one wants). The quality of the result should not depend on the frame size N.

Exercise 5.6.7 (Necessity of logarithmic oversampling)**.** ☕☕ Show that in general, logarithmic oversampling is necessary for covariance estimation. More precisely, give an example of a distribution in $\mathbb{R}^n$ for which the bound (5.20) must fail for every $\varepsilon < 1$ unless $m \gtrsim n \log n$.

Hint: Think about the coordinate distribution from Section 3.3.4; argue like in Exercise 5.4.14.

[17] The concept of frames was introduced in Section 3.3.4. By equal-norm frame we mean that $\|u_i\|_2 = \|u_j\|_2$ for all i and j.

Exercise 5.6.8 (Random matrices with general independent rows)**.** ☕☕☕ Prove a version of Theorem 4.6.1 which holds for random matrices with arbitrary, not necessarily sub-gaussian distributions of rows.

Let A be an $m \times n$ matrix whose rows A_i are independent isotropic random vectors in $\mathbb{R}^n$. Assume that for some $K \geq 0$,

$$\|A_i\|_2 \leq K\sqrt{n} \quad \text{almost surely for every } i. \tag{5.21}$$

Prove that, for every $t \geq 1$, one has

$$\sqrt{m} - Kt\sqrt{n \log n} \leq s_n(A) \leq s_1(A) \leq \sqrt{m} + Kt\sqrt{n \log n} \tag{5.22}$$

with probability at least $1 - 2n^{-ct^2}$.

Hint: Just like in the proof of Theorem 4.6.1, derive the conclusion from a bound on $\frac{1}{m}A^\mathsf{T}A - I_n = \frac{1}{m}\sum_{i=1}^m A_iA_i^\mathsf{T} - I_n$. Use the result of Exercise 5.6.4.

5.7 Notes

There are several introductory texts about concentration, such as [11, Chapter 3], [150, 130, 129, 30] and an elementary tutorial [13].

The approach to concentration via isoperimetric inequalities that we presented in Section 5.1 was fist discovered by P. Lévy, to whom Theorems 5.1.5 and 5.1.4 are due (see [91]).

When V. Milman realized the power and generality of Lévy's approach in 1970-s, this led to far-reaching extensions of the *concentration of measure* principle, some of which we surveyed in Section 5.2. To keep this book concise, we left out a lot of important approaches to concentration, including bounded differences inequality, martingale, semigroup and transportation methods, Poincare inequality, log-Sobolev inequality, hypercontractivity, Stein's method and Talagrand's concentration inequalities see [212, 129, 30]. Most of the material we covered in Sections 5.1 and 5.2 can be found in [11, Chapter 3], [150, 129].

The Gaussian isoperimetric inequality (Theorem 5.2.1) was first proved by V. N. Sudakov and B. S. Cirelson (Tsirelson) and independently by C. Borell [28]. There are several other proofs of Gaussian isoperimetric inequality, see [24, 12, 16]. There is also an elementary derivation of Gaussian concentration (Theorem 5.2.2) from Gaussian interpolation instead of isoperimetry, see [167].

Concentration on the Hamming cube (Theorem 5.2.5) is a consequence of Harper's theorem, which is an isoperimetric inequality for the Hamming cube [98], see [25]. Concentration on the symmetric group (Theorem 5.2.6) is due to B. Maurey [139]. Both Theorems 5.2.5 and 5.2.6 can be also proved using martingale methods, see [150, Chapter 7].

The proof of concentration on Riemannian manifolds with positive curvature (inequality [129, Section 2.3]) can be found e.g. in [129, Proposition 2.17]. Many interesting special cases follow from this general result, including Theorem 5.2.7 for the special orthogonal group [150, Section 6.5.1] and, consequently, Theorem 5.2.9 for the Grassmannian [150, Section 6.7.2]. A construction of Haar measure we mentioned in Remark 5.2.8 can be found e.g. in [150, Chapter 1] and

[76, Chapter 2]; the survey [147] discusses numerically stable ways to generate random unitary matrices.

Concentration on the continuous cube (Theorem 5.2.10) can be found in [129, Proposition 2.8], and concentration on the Euclidean ball (Theorem 5.2.13), in [129, Proposition 2.9]. Theorem 5.2.15 on concentration for exponential densities is borrowed from [129, Proposition 2.18]. The proof of Talagrand's concentration inequality (Theorem 5.2.16) originally can be found in [198, Theorem 6.6], [129, Corollary 4.10].

The original formulation of Johnson-Lindenstrauss Lemma is from [110]. For various versions of this lemma, related results, applications, and bibliographic notes, see [138, Section 15.2]. The condition $m \gtrsim \varepsilon^{-2} \log N$ is known to be optimal [124].

The approach to matrix concentration inequalities we follow in Section 5.4 originates in the work of R. Ahlswede and A. Winter [4]. A short proof of Golden-Thompson inequality (Theorem 5.4.7), a result on which Ahlswede-Winter's approach rests, can be found e.g. in [21, Theorem 9.3.7] and [221]. While the work of R. Ahlswede and A. Winter was motivated by problems of quantum information theory, the usefulness of their approach was gradually understood in other areas as well; the early work includes [227, 220, 92, 159].

The original argument of R. Ahlswede and A. Winter yields a version of matrix Bernstein's inequality that is somewhat weaker than Theorem 5.4.1, namely with $\sum_{i=1}^{N} \| \mathbb{E} X_i^2 \|$ instead of σ. This quantity was later tightened by R. Oliveira [160] by a modification of Ahlswede-Winter's method and independently by J. Tropp [206] using Lieb's inequality (Theorem 5.4.8) instead of Golden-Thompson's. In this book, we mainly follow J. Tropp's proof of Theorem 5.4.1. The book [207] presents a self-contained proof of Lieb's inequality (Theorem 5.4.8), matrix Hoeffding's inequality from Exercise 5.4.12, matrix Chernoff inequality, and much more. Due to J. Tropp's contributions, now there exist matrix analogs of almost all of the classical scalar concentration results [207]. The survey [165] discusses several other useful trace inequalities and outlines proofs of Golden-Thompson inequality (in Section 3) and Lieb's inequality (embedded in the proof of Proposition 7). The book [78] also contains a detailed exposition of matrix Bernstein's inequality and some of its variants (Section 8.5) and a proof of Lieb's inequality (Appendix B.6).

Instead of using matrix Bernstein's inequality, one can deduce the result of Exercise 5.4.11 from Gaussian integration by parts and a trace inequality [209]. Matrix Khintchine inequality from Exercise 5.4.13 can alternatively be deduced from *non-commutative Khintchine's inequality* due to F. Lust-Piquard [134]; see also [135, 40, 41, 172]. This derivation was first observed and used by M. Rudelson [175] who proved a version of Exercise 5.4.13.

For the problem of community detection in networks we discussed in Section 5.5, see the notes at the end of Chapter 4. The approach to concentration of random graphs using matrix Bernstein's inequality we outlined in Section 5.5 was first proposed by R. Oliveira [160].

In Section 5.6 we discussed covariance estimation for general high-dimensional distributions following [222]. An alternative and earlier approach to covariance estimation, which gives similar results, relies on matrix Khintchine's inequalities (known as non-commutative Khintchine inequalities); it was developed earlier by M. Rudelson [175]. For more references on covariance estimation problem, see the notes at the end of Chapter 4. The result of Exercise 5.6.8 is from [222, Section 5.4.2].